Report R8 — C150 at open boundaries: an exactly solved unitary QCA, and exactly what it leaves open

Krylov sector analysis of quantum cellular automata

QCA Sector Analysis project (Tier 1)

2026-07-26 (engine_version `1a.1`)

Abstract

C150 — the quantum elementary cellular automaton with Wolfram number 150, local symbol table (`I`, `V`, `V`, `I`), written `W150` elsewhere in this series — is the member of the 16-rule unitary family whose sector structure we can solve in closed form at open boundaries. We do so here, and then measure precisely how much of the dynamics that solution does *not* determine. Three results. (i) *Reduction.* For any unitary rule the sector partition equals the connected components of the single-flip graph $x \sim x \oplus 2^i$ over the sites at which the Hadamard fires; the brick-wall ordering drops out. Verified against the streamed engine on all 16 unitary rules, both boundary conventions, $N = 3 \dots 12$ (320 units, no mismatch). (ii) *Solution.* In the bond variables $b_j = x_j \oplus x_{j+1}$ the map is a bijection onto the even-weight strings of length $N + 1$ and the elementary move is a hard-core *hop* of one domain wall. Hence the conserved charge is the open Ising domain-wall number $Q = \sum_j (1 - Z_j Z_{j+1})/2$, the sectors are its eigenspaces, $|S_w| = \binom{N+1}{w}$ for even w, the sector count is $\lfloor (N+1)/2 \rfloor + 1$ and $D_{\max} = \max_{w\,\text{even}} \binom{N+1}{w}$. Because the sector count is *linear* in N, C150 is symmetry-resolved and not fragmented: it is the natural null model for the family. The exponential base of $D_{\max}$ is exactly 2 with an $N^{-1/2}$ prefactor, which supersedes the fitted 2.022 of R2. We push the assumption-free engine to $N = 22$ (10.2 hours for that unit alone) and the reduction to $N = 32$ — 4.29×10^9 basis states, 21 GB, 815 s, the RAM ceiling. Where the two overlap the reduction is 10^5 times faster and gives the identical size multiset, and the closed form is exact at every N either route reaches. (iii) *What is left open.* The sector is the maximal reachable set, not the dynamics. U has an exponentially large stationary space, $\dim \mathrm{Fix}(U_w) = \binom{\lceil N/2 \rceil}{w/2}$ and $\dim \mathrm{Fix}(U) = 2^{\lceil N/2 \rceil}$ (an exact law, verified two independent ways), almost all of it *dark superpositions* that no basis state names. The spectrum of U_w is degenerate for *every* shell with $|S_w| > N + 1$ and nondegenerate for the extremal ones, so outside those extremes $\dim \mathcal{K}(x) < |S_w|$ without exception: an inner sector is never the Krylov space of one of its basis states. Under monitoring the sector *is* filled uniformly (provably); without monitoring the diagonal ensemble occupies between 6.6% and 50% of it. (iv) *Interacting, but not chaotic.* In the bond representation C150 is a free-fermion Hadamard wall walk dressed by exactly two diagonal signs — a maximal nearest-neighbour wall interaction and a Jordan–Wigner string on the coin — and removing either one alone leaves the many-body spectrum non-additive while removing both makes it additive to 4×10^{-15}: C150 is genuinely interacting. It is nevertheless *not* a random-matrix system, and the reason is a hard count rather than a statistic. The number of distinct eigenvalues of the one-cycle operator on the full 2^N space is exactly 3^m for even N and $(3^m + (-1)^m)/2$ for odd N, with $m = \lceil N/2 \rceil$ the number of even sites — verified for $N = 4 \dots 13$ and then out of sample at $N = 14$, where the law predicts 2187 and the 16 384-dimensional diagonalisation returns 2187. The distinct fraction therefore decays as $(\sqrt{3}/2)^N = 0.866^N$, and no chaotic Floquet operator has an exponentially degenerate spectrum with a closed-form count. There is more structure still: for each N a *single* universal set Θ_N of that size carries every shell's spectrum as a subset, and the largest shell — the one realising $D_{\max}$ — realises *all* of Θ_N while the small shells realise as little as 1.8%. So the spectral problem is "find 3^m angles and decide which shell takes which", and that is not a random point process of any kind: not $\beta = 2$ (excluded by time reversal), not $\beta = 1$ (excluded by the degeneracy law), and not

Poisson either. The spacing statistics agree: measured against COE, Haar $O(D)$, CUE and Poisson references generated through the identical pipeline, the dilute shells sit at Poisson (as few-body shells must) and at half filling the reflection-resolved $\langle\tilde{r}\rangle$ falls from 0.60/0.55 at $N = 11$ to 0.37 at $N = 15$ and 0.35 at $N = 16$ — down to Poisson and past it, not up toward an RMT value — and neither superposition nor near-degeneracy can account for going below Poisson, which is the point. $\beta = 2$ (GUE/CUE) is excluded on symmetry grounds in any case: U is real orthogonal, so complex conjugation is a time reversal. The structural source of all of this is that $U = L_B L_A$ is a product of two commuting-gate *involutions*, which also explains the $\pm\theta$ pairing and the fixed space above.

Contents

1 Why this rule, and a naming note

Throughout this series rules are named by Wolfram number with a `W` prefix (`W150`); the present report follows the user-facing shorthand *C150* for the same object. The encoding is the standard one of Tier 1: the Wolfram number is read as a 4-tuple of local symbols indexed by the control pattern (x_{i-1}, x_{i+1}), with `I` the identity and `V` the Hadamard, so

$$150 \longmapsto (\mathtt{I}, \mathtt{V}, \mathtt{V}, \mathtt{I}), \qquad \text{i.e.} \qquad \text{Hadamard fires at site } i \iff x_{i-1} \oplus x_{i+1} = 1. \tag{1}$$

The firing condition is exactly the update of the *classical* additive elementary rule 90; C150 is the automaton that applies a Hadamard wherever ECA 90 would write a one. One cycle is the brick-wall product of a layer on the even sites followed by a layer on the odd sites, controls always read from the current bitmask. Boundaries: `obc0` fixes $x_{-1} = x_N = 0$, which is the case treated here; the ring `pbc` appears only for contrast in §4.

C150 is singled out for three reasons. It is one of the two rules (150 and 105) whose sector sizes R2 identified as binomials, so a closed form was plausible; it is the rule whose `obc0` sector sizes are used as a regression fixture against the external HSF oracle in R1; and its fitted growth exponent (2.022) sat awkwardly above the informational ceiling 2, which suggested that the fit was seeing a prefactor rather than a base. All three are settled below.

2 Reduction: the sector partition of a unitary rule is a flip graph

Theorem 1 (flip reduction)**.** *Let the rule be unitary (symbols in* $\{\mathtt{I}, \mathtt{V}\}$*) and let* $s_i(x) = t[2x_{i-1} + x_{i+1}]$ *be the symbol fired at site i when the controls read x. Then the sector partition — the weakly connected components of the one-cycle transition graph, which for a unitary rule coincide with its strongly connected components — equals the connected components of the undirected graph*

$$F: \quad x \sim x \oplus 2^i \qquad \textit{for every } i \textit{ with } s_i(x) = \mathtt{V}. \tag{2}$$

Proof. A cycle is two depth-1 half-layers. By the no-interference theorem of R1, each half-layer is — conditioned on the frozen complementary sublattice — a product of single-qubit gates acting on disjoint qubits, so the image of a basis state x under one half-layer is exactly the *subcube* spanned by the sites whose symbol is `V`, with the complementary sublattice held fixed. Two consequences. First, x lies in its own half-layer image, because $\langle b|\mathtt{I}|b\rangle = 1$ and $\langle b|H|b\rangle = \pm 1/\sqrt{2}$ are both nonzero; hence the composite image $\Phi(x)$ contains both half-layer images $A(x)$ and $B(x)$. Second, inside a half-layer image the frozen sublattice — and therefore the `V`/`I` pattern itself — does not change, so the subcube is connected by single flips which are themselves edges of F. By the first point every F-edge is realised by a composite path; by the second, every composite edge $x \to z$ (with $z \in B(y)$, $y \in A(x)$) is realised by the F-path $x \sim y \sim z$. The two graphs have the same components. □

Two remarks. The reduction is *layer-agnostic*: the brick-wall ordering disappears, because the controls of site i are read off x itself in the half-layer in which i is frozen. The sector partition of a unitary QECA is therefore a property of the local symbol table alone — not of the circuit schedule. And the cost collapses from $O(2^N|\text{succ}|)$, with $|\text{succ}|$ growing exponentially for the Hadamard-rich rules, to $O(N2^N)$ with a tiny constant; §5 uses this.

Validation. Theorem 1 is checked against the streamed engine (`graph.scc.analyze`, which knows nothing about it) for all 16 unitary rules, both boundary conventions and $N = 3 \ldots 12$: 320 units, full size multisets compared, no mismatch, for both the pure-python union-find and the vectorised label-propagation implementation. See `tests/test_flip_graph.py`.

3 Solution: walls, and the Ising charge

Introduce the bond (domain-wall) variables. For `obc0` there are $N+1$ bonds,

$$b_j = x_j \oplus x_{j+1}, \quad j = 0, \ldots, N, \qquad x_{-1} = x_N = 0, \tag{3}$$

with the index convention of the code ($b_0 = x_0$, $b_N = x_{N-1}$).

Proposition 1 (bond bijection). *$x \mapsto b$ is a bijection from $\{0,1\}^N$ onto the even-weight strings of length $N+1$; the inverse is the prefix XOR $x_j = \bigoplus_{k\le j} b_k$. On the ring the same map is exactly 2-to-1 onto the even-weight strings of length N, the two preimages being x and its global spin flip.*

Both statements are verified exhaustively for $N = 2 \ldots 12$ (obc0) and $N = 3 \ldots 12$ (pbc).

Now translate the flip move (2). Flipping x_i toggles exactly the two bonds b_i and b_{i+1}, and by (1) the flip is allowed iff $x_{i-1} \neq x_{i+1}$, i.e. iff $b_i \neq b_{i+1}$. So the elementary move is

$$(b_i, b_{i+1}) = (1,0) \;\longleftrightarrow\; (0,1): \qquad \text{a } \textit{hard-core hop} \text{ of one domain wall.} \tag{4}$$

Creation and annihilation of walls is forbidden — the configuration $b_i = b_{i+1}$ is precisely where the symbol is `I`. Therefore:

Corollary 1 (conserved charge). *The wall number $w = \sum_j b_j$ is conserved exactly, i.e. $[U, Q] = 0$ for the strictly 2-local diagonal operator*

$$Q \;=\; \sum_{j=0}^{N} \frac{1 - Z_j Z_{j+1}}{2}, \qquad Z_{-1} = Z_N = +1, \tag{5}$$

the open Ising domain-wall number with boundary fields. (Q is diagonal, so conservation at the level of supports already gives the operator identity; support-level conservation is verified for every x, $N = 3 \ldots 12$, both boundary conventions, with zero violations.)

Corollary 2 (connectivity). *Each wall shell is one sector. Hard-core hopping on a segment of $N+1$ sites is connected on every fixed-occupancy shell (bubble the walls to the leftmost packed configuration), and (4) is exactly that dynamics.*

4 Closed form, and the growth law

Combining Proposition 1 with Corollaries 1 and 2:

$$\boxed{|S_w| = \binom{N+1}{w} \;\; (w \text{ even}), \qquad \#\text{sectors} = \left\lfloor \frac{N+1}{2} \right\rfloor + 1, \qquad D_{\max} = \max_{w \text{ even}} \binom{N+1}{w}.} \tag{6}$$

The frozen sectors are the singletons $w = 0$ (all-zero) and, when N is odd, $w = N+1$ (the alternating configuration); for even N there is only one. The parity bookkeeping in $D_{\max}$ is not cosmetic: for $N \equiv 1 \pmod 4$ the central binomial sits at odd w and is *not* a sector size, so $D_{\max}$ is the near-central one (e.g. $N = 17$: $\binom{18}{8} = 43758$, not $\binom{18}{9}$).

Ring, for contrast. On the ring the wall map is 2-to-1, so a shell with $0 < w < N$ lifts to one sector of size $2\binom{N}{w}$, while the frozen shells $w = 0$ and (for even N) $w = N$ each split into *two* singletons — no hop is available, and the two spin preimages are disconnected. Hence $\#\text{sectors} = N/2 + 3$ for even N and $(N+3)/2$ for odd N. Both closed forms reproduce the stored Tier-1a data at every $N = 6 \ldots 20$, in both boundary conventions, size multiset for size multiset.

C150 is not fragmented. Equation (6) has a sector count that is *linear* in N and sector sizes that are full eigenspaces of the single charge Q. There is no Krylov structure beyond the $U(1)$ symmetry: C150 is *symmetry-resolved*, not fragmented. This is worth stating plainly because C150 is the rule most often quoted from this family, and it is exactly the wrong example for fragmentation. The fragmented members of the family look nothing like it — `W204` has 2^N frozen singletons, and the wall families of `W51`/`W201` carry exponentially many sectors. C150 is the family's null model.

The growth law: base exactly 2, prefactor $N^{-1/2}$. Since $\sum_{w\,\text{even}} \binom{N+1}{w} = 2^N$ and the maximum is (near-) central,

$$\frac{D_{\max}}{2^N} = 2\sqrt{\frac{2}{\pi(N+1)}}\left(1+O(N^{-1})\right) \longrightarrow 0, \tag{7}$$

so $D_{\max}$ grows as $2^N N^{-1/2}$: the exponential base is *exactly* 2 and the sub-exponential correction is a square root. The approach is not smooth but a sawtooth (Figure 1, right), which is the parity bookkeeping above: for $N \equiv 1 \pmod 4$ the true central binomial is at odd w and $D_{\max}$ has to settle for the neighbour, which costs a factor $1 - O(N^{-1})$ and produces the dips at $N = 5, 9, 13, \ldots$. A pure-exponential fit sees this sawtooth on top of the $N^{-1/2}$ decay, which is exactly the configuration that pushes an unconstrained base slightly above the true one. The value 2.022 reported for C150/`obc0` in R2's pure-exponential fit is the finite-size shadow of that prefactor, not a base above the informational ceiling; R2's companion observation that 150/105 carry $\alpha = -\frac{1}{2}$ is the same statement seen from the fit side, and (7) now supplies it exactly. Figure 1 (right) overlays the exact ratio on (7).

5 Numerical frontier

Two frontiers are reported, deliberately kept separate.

- The **assumption-free frontier**: the streamed engine (`graph.scc.sectors_union_find` over `core.cycle.succ`), which propagates exact $\mathbb{Z}[1/\sqrt{2}]$ amplitudes and uses no result of this report. Its cost grows by a factor ≈ 3 per site because the one-cycle support itself grows exponentially, so it is the binding runtime constraint: $N = 20$ already takes 65 minutes.
- The **reduction frontier**: Theorem 1 implemented as vectorised label propagation on the flip graph, with the label array (`uint32`, 4 bytes per basis state) the only object of size 2^N — pointer jumping is done in place and in chunks, and the size histogram is harvested chunkwise, so nothing else of that size is ever allocated. This is RAM-bound rather than time-bound, and it reaches $N = 32$: $2^{32} = 4\,294\,967\,296$ basis states, 17 sectors, $D_{\max} = 1\,166\,803\,110$, in 815 s at a peak of 21.0 GB. That is the ceiling twice over — $N = 32$ is the largest index a `uint32` label can hold, and $N = 33$ would need a 64-bit label array of 68.7 GB.

Both agree with each other and with (6) at every N computed. Table 1 lists the units; runtimes are wall-clock on the project machine (20 cores, 31 GB, jobs `nice`d and run concurrently). The two routes overlap at $N = 21$ and $N = 22$, where they return the identical size multiset and the reduction is faster by 4.0×10^4 and 1.0×10^5 respectively (13 018 s versus 0.32 s, 36 728 s versus 0.37 s). That ratio is the price of streaming the full one-cycle support instead of a spanning set of edges, and it is what makes the difference between $N = 22$ and $N = 32$.

N	2^N	#sectors	$D_{\max}$	$D_{\max}/2^N$	engine	flip graph
14	16 384	8	6 435	0.3928	✓ 4.4 s	—
15	32 768	9	12 870	0.3928	✓ 14 s	—
16	65 536	9	24 310	0.3709	✓ 40 s	—
17	131 072	10	43 758	0.3338	✓ 126 s	—
18	262 144	10	92 378	0.3524	✓ 363 s	—
19	524 288	11	184 756	0.3524	✓ 1321 s	—
20	1 048 576	11	352 716	0.3364	✓ 1.1 h	—
21	2 097 152	12	646 646	0.3083	✓ 3.6 h	✓ 0.3 s
22	4 194 304	12	1 352 078	0.3224	✓ 10.2 h	✓ 0.4 s
23	8 388 608	13	2 704 156	0.3224	—	✓ 0.9 s
24	16 777 216	13	5 200 300	0.3100	—	✓ 1.8 s
25	33 554 432	14	9 657 700	0.2878	—	✓ 3.7 s
26	67 108 864	14	20 058 300	0.2989	—	✓ 8.9 s
27	134 217 728	15	40 116 600	0.2989	—	✓ 18 s
28	268 435 456	15	77 558 760	0.2889	—	✓ 38 s
29	536 870 912	16	145 422 675	0.2709	—	✓ 72 s
30	1 073 741 824	16	300 540 195	0.2799	—	✓ 154 s
31	2 147 483 648	17	601 080 390	0.2799	—	✓ 297 s
32	4 294 967 296	17	1 166 803 110	0.2717	—	✓ 815 s

Table 1: C150 `obc0`: exact sector data and the two numerical frontiers. ✓ means the computed size multiset equals the closed form (6) entry for entry; the time is that unit's wall clock; "—" means that route has not been run at that N (the engine is a factor ≈ 3 per site, so its units above $N = 20$ are hours to days each). The closed-form columns are exact for every N, computed or not.

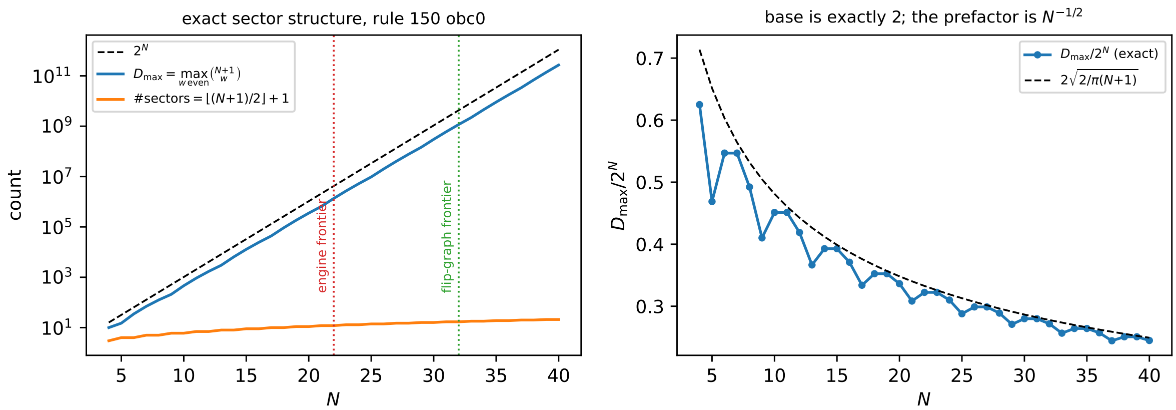

Figure 1: Left: the exact sector structure (6) with the two computed frontiers marked. Right: $D_{\max}/2^N$ against the asymptotic $2\sqrt{2/\pi(N+1)}$ of (7) — the base is 2 and the decay is the $N^{-1/2}$ prefactor that a pure-exponential fit misreads as 2.022.

6 What the sector does *not* determine

A sector is the minimal U-invariant subspace spanned by basis states: the *maximal kinematically reachable* set from any of its members, and nothing more (R1, §"What a sector is"). For C150 we can now quantify the gap exactly, because the shells are small enough to diagonalise. The dataset is every shell of every $N = 4 \dots 24$ with $|S_w| \le 3000$: 70 shells, the largest of dimension 2380 ($N = 16$, $w = 4$). All numbers below come from U_w built column by column from the engine's exact amplitudes and verified orthogonal to 10^{-9}. The eigenvalue clustering tolerance is 10^{-9}, and it has margin in both directions: across the whole dataset the degeneracies of U_w are exact (largest gap treated as a degeneracy 8.8×10^{-15}, five orders below the tolerance) while the smallest gap treated as *distinct* is 4.8×10^{-6}, more than three orders above it.

An exponentially large stationary space. U does not merely fail to be ergodic inside a sector; it has a huge kernel of $U - \mathbb{1}$:

$$\boxed{\dim \mathrm{Fix}(U_w) \;=\; \binom{\lceil N/2 \rceil}{w/2}, \qquad \dim \mathrm{Fix}(U) \;=\; 2^{\lceil N/2 \rceil} \;=\; 2^{\#\{\text{even sites}\}}.} \tag{8}$$

The shell law is verified over the whole dataset — every shell of every $N = 4 \dots 24$ with $|S_w| \le 3000$, 70 shells, no mismatch; the total is verified for $N = 3 \dots 12$ by two independent routes, the multiplicity of the eigenvalue $+1$ and $\dim \ker(U - \mathbb{1})$ by rank. The mechanism is visible in the half-layer structure: writing $U = L_B L_A$ with L_A, L_B the (Hermitian, involutive) half-layers, one finds *numerically for* $N = 3 \dots 11$ that

$$\mathrm{Fix}(U) \;=\; \mathrm{Fix}(L_A) \cap \mathrm{Fix}(L_B), \tag{9}$$

i.e. a state stationary under the cycle is already stationary under each half-layer separately, and each half-layer is a product of commuting single-qubit involutions (H or $\mathbb{1}$ per site of its sublattice), each of which contributes a one-dimensional local $+1$ eigenspace. The counting exponent $\lceil N/2 \rceil$ is the number of even sites, i.e. the width of the first half-layer. Only the $w = 0$ (and, for odd N, $w = N + 1$) members of $\mathrm{Fix}(U)$ are basis states; *all the rest are dark superpositions*, invisible to any computational-basis graph.

Consequence: an inner sector is never a Krylov space. For a basis state $|x\rangle$, $\dim \mathcal{K}(x)$ equals the number of distinct eigenvalues of U_w that $|x\rangle$ overlaps, so any degeneracy bounds it strictly below $|S_w|$. Over the whole dataset the dividing line is sharp and depends only on the sector dimension:

$$\begin{aligned} \text{spectrum of } U_w \text{ nondegenerate} \quad &\Longleftrightarrow \quad |S_w| \le N + 1 \\ &\Longleftrightarrow \quad w \in \{0,\, N+1\} \quad \text{or} \quad (N \text{ even and } w = N), \end{aligned} \tag{10}$$

i.e. exactly the extremal shells — the two frozen singletons and, for even N, the top shell of dimension $N + 1$, where (8) gives $\dim \mathrm{Fix}(U_w) = 1$ and no further multiplicity appears. For every one of the 59 *inner* shells ($|S_w| > N + 1$) the spectrum is degenerate and

$$\dim \mathcal{K}(x) \;<\; |S_w|, \qquad \frac{\dim \mathcal{K}(x)}{|S_w|} \in [0.395,\, 0.963], \tag{11}$$

the fraction decreasing with sector size (Table 2, Figure 2). This is the sharp form, for this rule, of the general caveat in R1: the graph gives $\dim \mathcal{K}(x) \le |S_w|$, and away from the extremal shells the inequality is always strict. For odd N the reflection P commutes with U exactly ($[P, U] = 0$ to machine zero, 30 of the 70 shells) and reflection-symmetric seeds lose roughly half their Krylov space again ($N = 7$, $w = 2$: 13 instead of 25) — the symmetry-forced degeneracy anticipated in R1, here measured. For even N the brick-wall order maps the sublattices into each other and reflection acts instead as a time reversal, $PUP = U^{-1}$ exactly (the other 40 shells).

Monitored versus unmonitored, made numerical. Under monitoring the sector *is* filled uniformly, provably: $M_w = |U_w|^2$ is doubly stochastic, irreducible on S_w (Corollary 2) and aperiodic (by the R1 no-interference theorem $\langle x|U|x\rangle \neq 0$ for every x, so every node carries a self-loop), hence $M_w^t \to$ uniform on S_w. Without monitoring the long-time state is the diagonal ensemble $\bar\rho = \sum_\lambda P_\lambda |x\rangle\langle x| P_\lambda$, and it is far from uniform: over the 59 inner shells its effective dimension $d_{\text{eff}} = 1/\operatorname{tr}\bar\rho^2$ occupies between 6.6% and 50% of the sector — never more than half — and the most-occupied basis state carries 2.0 to 12.5 times the uniform weight (Table 2). So for C150 the two experiments genuinely disagree about the long-time state *inside* a sector, even though they agree exactly about which sector that state lives in, and (by R1) about populations after a single cycle.

N	w	$\lvert S_w\rvert$	#spec	dim Fix	$\binom{\lceil N/2\rceil}{w/2}$	$\dim\mathcal{K}/\lvert S_w\rvert$	$d_{\text{eff}}/\lvert S_w\rvert$	$\max p\cdot\lvert S_w\rvert$	reflection
8	2	36	33	4	4	0.917	0.388	2.57	$PUP = U^{-1}$
8	4	126	81	6	6	0.643	0.222	4.46	$PUP = U^{-1}$
8	6	84	65	4	4	0.774	0.292	3.42	$PUP = U^{-1}$
10	2	55	51	5	5	0.927	0.359	2.78	$PUP = U^{-1}$
10	4	330	211	10	10	0.639	0.156	6.33	$PUP = U^{-1}$
10	6	462	243	10	10	0.526	0.137	7.17	$PUP = U^{-1}$
10	8	165	131	5	5	0.794	0.241	4.12	$PUP = U^{-1}$
10	$10^\dagger$	11	11	1	1	1.000	0.793	1.26	$PUP = U^{-1}$
12	2	78	73	6	6	0.936	0.337	2.96	$PUP = U^{-1}$
12	4	715	473	15	15	0.662	0.123	8.07	$PUP = U^{-1}$
12	6	1 716	729	20	20	0.425	0.079	12.49	$PUP = U^{-1}$
12	8	1 287	665	15	15	0.517	0.096	10.15	$PUP = U^{-1}$
12	10	286	233	6	6	0.815	0.218	4.58	$PUP = U^{-1}$
12	$12^\dagger$	13	13	1	1	1.000	0.785	1.27	$PUP = U^{-1}$
14	2	105	99	7	7	0.943	0.316	3.13	$PUP = U^{-1}$
14	4	1 365	939	21	21	0.688	0.106	9.37	$PUP = U^{-1}$
14	12	455	379	7	7	0.833	0.201	4.98	$PUP = U^{-1}$
14	$14^\dagger$	15	15	1	1	1.000	0.779	1.28	$PUP = U^{-1}$
16	2	136	129	8	8	0.949	0.312	3.20	$PUP = U^{-1}$
16	4	2 380	1 697	28	28	0.713	0.093	10.75	$PUP = U^{-1}$
16	14	680	577	8	8	0.849	0.191	5.08	$PUP = U^{-1}$
16	$16^\dagger$	17	17	1	1	1.000	0.774	1.29	$PUP = U^{-1}$

Table 2: Wall shells of C150 `obc0`. #spec is the number of distinct eigenvalues of U_w; dim Fix its $+1$ multiplicity, beside the closed form (8); $\dim\mathcal{K}/|S_w|$ the largest Krylov fraction over the sampled seeds; $d_{\text{eff}}/|S_w|$ the smallest diagonal-ensemble fraction; $\max p\cdot|S_w|$ the peak long-time population in units of the uniform value (1 would be uniform, which is what monitoring gives exactly).

7 C150 is charge-resolved but interacting

The wall picture invites the guess that C150 is a free-fermion circuit. It is not, and the obstruction is exactly two signs.

Write the cycle on the $N+1$ bond modes. The gate at site i touches bonds i and $i+1$ only; it is the identity when both carry the same occupation, and on the one-wall subspace — basis ordered (wall at bond i, wall at bond $i+1$) — it is

$$M(c) \;=\; \frac{1}{\sqrt{2}}\begin{pmatrix} -(-1)^c & 1 \\ 1 & (-1)^c \end{pmatrix}, \qquad \det M(c) = -1, \tag{12}$$

where $c = x_{i-1} = \bigoplus_{k<i} b_k$ is the parity of the walls strictly to the left of bond i — a Jordan–Wigner string. This bond-space circuit reproduces the engine's U under the bijection of Proposition 1 to 1.1×10^{-16} (Table 3, second column), so (12) *is* C150: a hard-core quantum walk of domain walls with a Hadamard coin.

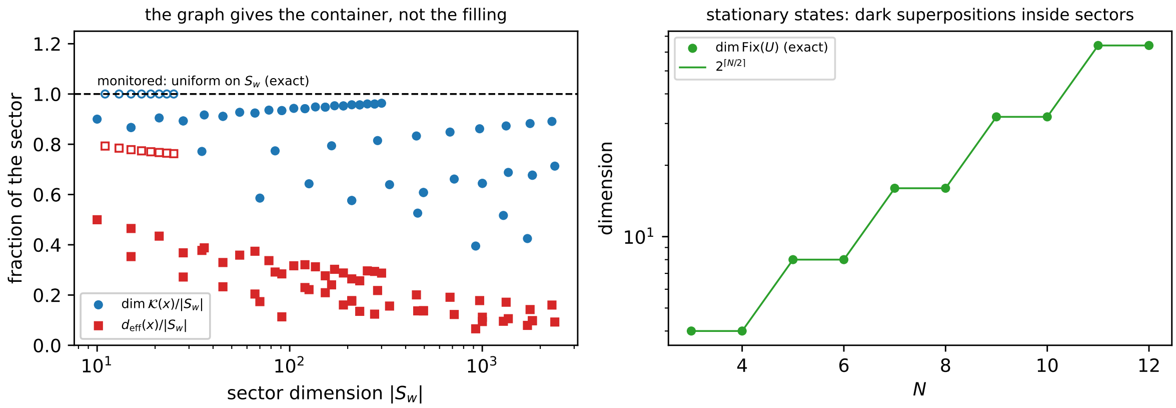

Figure 2: Left: the Krylov and diagonal-ensemble fractions of a sector against sector dimension; the dashed line is the monitored answer, which is exactly uniform. Filled markers are inner shells; open markers the extremal shells $|S_w| \le N+1$ — the only ones with a nondegenerate spectrum, and hence the only ones a basis state's Krylov space fills. Right: the stationary space $\dim \mathrm{Fix}(U) = 2^{\lceil N/2 \rceil}$, whose members other than the frozen basis states are dark superpositions.

Two features of (12) spoil gaussianity. A number-conserving Gaussian two-mode gate with single-particle block u must act on the doubly occupied state by $\det u$; here $\det M = -1$ while the gate acts by $+1$, which is a maximal nearest-neighbour interaction $\exp(i\pi n_i n_{i+1})$ between walls. And the $(-1)^c$ string multiplies the *density-difference* part of the coin, not the hop, so it is not absorbed by the Jordan–Wigner transform. The test is sharp: a number-conserving Gaussian circuit acts on the w-particle sector as the w-th antisymmetric power of its single-particle block, so its eigenphases are subset sums of the one-particle phases. Table 3 reports the largest failure of that additivity over $w = 2 \ldots 5$ for the four sign variants.

N	bond vs engine	rule 150	no string	det phase only	Gaussian ref.
5	1.1×10^{-16}	1.871	1.101	1.340	8.9×10^{-16}
6	1.1×10^{-16}	1.157	0.869	0.869	1.4×10^{-15}
7	1.1×10^{-16}	1.062	0.608	0.526	1.4×10^{-15}
8	1.1×10^{-16}	0.731	0.688	0.515	3.0×10^{-15}
9	1.1×10^{-16}	0.549	0.604	0.427	3.7×10^{-15}

Table 3: Additivity defect: the largest $|\arg(\lambda/\lambda')|$ in radians between the w-wall spectrum and the w-fold products of the one-wall eigenvalues, maximised over $w = 2 \ldots 5$ (compared as points on the unit circle, so the branch cut cannot masquerade as a defect). Column 2 is the bond representation checked against the engine. C150 (column 3) and each of the two single-defect models (columns 4, 5) are non-additive at 0.43–1.87 rad; removing *both* defects gives the Gaussian reference (column 6), additive to 4×10^{-15}.

The conclusion is a clean separation of the two questions. The *fragmentation* of C150 is entirely a $U(1)$ symmetry and is solved in closed form; the *dynamics within a sector* is that of interacting hard-core walls, and the interaction is not a perturbation — it is maximal. That is the honest reason a sector's spectral properties do not follow from (6), and why (8) had to be measured rather than read off the graph.

8 Level statistics: interacting, but not chaotic

8.1 Why the comparison is not to GUE

Two exact structures fix the symmetry class before any number is computed.

U is real orthogonal. Every amplitude lies in $\mathbb{Z}[1/\sqrt{2}]$, so complex conjugation K is an antiunitary symmetry, $KUK = U$ with $K^2 = 1$. That is time-reversal invariance: the $\beta = 2$ ensembles (GUE, CUE) are excluded on symmetry grounds, and the candidates are the $\beta = 1$ family. It also means the spectrum is closed under conjugation, so eigenvalues come in mirror pairs $e^{\pm i\theta}$ together with real eigenvalues ± 1. That mirror is a kinematic constraint, not a spectral correlation, so every statistic below is computed on the *independent* half of the spectrum, $\theta \in (0, \pi)$, with ± 1 removed.

U is a product of two involutions. Each half-layer is a product of commuting single-site gates, and each such gate is a real symmetric involution — on the doubly and doubly-un-occupied bond pairs it is the identity, and on the one-wall subspace $M(c)^2 = \nVdash$ because $\operatorname{tr} M(c) = 0$ and $\det M(c) = -1$ in (12). So

$$U = L_B L_A, \qquad L_A = L_A^{\mathsf{T}} = L_A^{-1}, \quad L_B = L_B^{\mathsf{T}} = L_B^{-1}, \qquad L_A U L_A = U^{-1}, \tag{13}$$

all verified to 10^{-15} for $N = 3 \dots 10$. This is a dihedral structure: on the 2-dimensional irreps of the algebra generated by L_A and L_B, U acts as a rotation by some angle θ; on the 1-dimensional ones $U = L_B L_A = \pm 1$. It also *explains* (8), which §6 could only report: $U\psi = \psi$ forces $L_A \psi = L_B \psi$, so

$$\operatorname{Fix}(U) = \big[\operatorname{Fix}(L_A) \cap \operatorname{Fix}(L_B)\big] \oplus \big[\operatorname{Fix}(-L_A) \cap \operatorname{Fix}(-L_B)\big], \tag{14}$$

and the second summand is *empty* for every $N = 3 \dots 11$, which is why $\operatorname{Fix}(U)$ is exactly the intersection of the two half-layer fixed spaces.

8.2 Pipeline and references

Each exact degeneracy is collapsed to one level: a spacing statistic run over a degenerate spectrum measures the degeneracies. The primary statistic is the spacing-ratio average $\langle \tilde{r} \rangle$ with $\tilde{r}_i = \min(r_i, 1/r_i)$, $r_i = s_i/s_{i-1}$, which needs no unfolding and so cannot be biased by a bad unfolding of a Floquet spectrum; the unfolded spacing distribution $P(s)$ (local moving-average unfolding) is reported beside it. Reference ensembles are *generated* and pushed through the identical pipeline rather than quoted from surmise constants, so the half-circle restriction and the degeneracy collapse affect data and reference equally: Poisson, COE ($U = W^{\mathsf{T}} W$, W Haar CUE), Haar $O(D)$ — the ensemble that shares C150's conjugate-pairing constraint — CUE, and superpositions of m independent Haar-O spectra, which is what *unresolved* symmetry blocks look like.

Two further exact facts organise the data. For odd N the reflection P commutes with U, so the shell is a superposition of two parity blocks and must be analysed inside each; superposition can only push a repulsion measure *down* toward Poisson, so unresolved blocks bias $\langle \tilde{r} \rangle$ low and never high. And for odd N the shells w and $N + 1 - w$ have *identical* spectra (verified to 1.2×10^{-14}; both weights are even because $N + 1$ is even), so they are counted once. Finally, filling matters: a shell with fixed w and growing N is w hard-core walls on $N + 1$ bonds, i.e. a *few-body* problem, which must approach Poisson however chaotic the dense model is. Only $\nu = w/(N+1)$ near $\frac{1}{2}$ can speak to many-body chaos.

8.3 The decisive fact: the spectrum is exponentially degenerate

Before the spacings, count the levels. The number of *distinct* eigenvalues of U on the whole 2^N space is not generic — it is exactly

$$\boxed{\#\{\text{distinct eigenvalues}\} \;=\; \begin{cases} 3^m, & N \text{ even}, \\ \big(3^m + (-1)^m\big)/2, & N \text{ odd}, \end{cases} \qquad m = \#\{\text{even sites}\} = \lceil N/2 \rceil.} \tag{15}$$

Verified for $N = 4 \dots 13$ (Table 4) and then *out of sample* at $N = 14$: the law was written down from $N \leq 13$, it predicts $3^7 = 2187$, and the 16 384-dimensional diagonalisation returns exactly 2187. So the distinct spectrum grows like $(\sqrt{3})^N$ while the Hilbert space grows like 2^N: the distinct fraction decays as $(\sqrt{3}/2)^N = 0.866^N$, and by $N = 14$ only 2187 of 16 384 eigenvalues are distinct.

The count does not depend on a tolerance. A degeneracy count is only as good as the threshold that defines it, so it is worth recording that here there is no threshold to choose. At $N = 12$ the 4095 consecutive phase gaps split into 3367 below 10^{-12} and 728 above 10^{-4}, with *nothing at all* in between — five empty decades separating the degeneracies from the distinct levels. Accordingly the count is unchanged for every tolerance from 10^{-13} to 10^{-5}, eight orders of magnitude, at $N = 8, 9, 10, 11, 12$ alike. Equation (15) is a property of the operator, not of the numerics.

This settles the question posed in §7 more sharply than any spacing statistic could, and it settles it the other way. *No chaotic Floquet operator has an exponentially degenerate spectrum obeying a closed-form count.* C150 is therefore not in the COE class, nor in any random-matrix class: it is an exactly structured operator. Combined with §7, the characterisation is that C150 is *interacting but not chaotic* — and there is no contradiction, because interacting and integrable are perfectly compatible (every Bethe-ansatz-solvable model is both). The base 3 per even site is the natural counting for the dihedral structure (13): each single-site gate is an involution with a 3-dimensional +1 eigenspace and a 1-dimensional −1 eigenspace on the bond pair it touches.

N	2^N	#distinct	closed form	value	fraction	
4	16	9	3^2	9	0.5625	✓
5	32	13	$(3^3-1)/2$	13	0.4062	✓
6	64	27	3^3	27	0.4219	✓
7	128	41	$(3^4+1)/2$	41	0.3203	✓
8	256	81	3^4	81	0.3164	✓
9	512	121	$(3^5-1)/2$	121	0.2363	✓
10	1 024	243	3^5	243	0.2373	✓
11	2 048	365	$(3^6+1)/2$	365	0.1782	✓
12	4 096	729	3^6	729	0.1780	✓
13	8 192	1 093	$(3^7-1)/2$	1 093	0.1334	✓
14	16 384	2 187	3^7	2 187	0.1335	✓

Table 4: The distinct-eigenvalue count of the full-space operator against (15). The last column is #distinct$/2^N$, which decays as 0.866^N.

Erratum to §7. The outlook of the previous version of this report predicted that the level statistics “will not look free”. Read as “will look chaotic” that prediction is wrong, and (15) is why: non-gaussianity of the gate rules out a *free-fermion* description, but it does not imply random-matrix behaviour, and here it does not produce it.

8.4 And the spacings agree

Table 6 and Figure 3 give $\langle\tilde{r}\rangle$ shell by shell against the generated references. The reading is consistent with (15) rather than with any RMT class: the dilute shells sit at Poisson, as they must, and the shells at half filling drift *downward* with N — the opposite of the flow toward an RMT value that a chaotic system shows. The cleanest sequence is exact half filling, $\nu = w/(N+1) = \frac{1}{2}$, with reflection resolved:

$$N = 11,\ w = 6:\ \langle\tilde{r}\rangle = 0.597,\ 0.550; \qquad N = 15,\ w = 8:\ \langle\tilde{r}\rangle = 0.366,\ 0.375, \tag{16}$$

i.e. from between COE and Haar-O down to *at* Poisson (0.384 ± 0.017 from our own generation) as N grows, while the degeneracy fraction of the same blocks rises from 0.61 to 0.75 — exactly what (15) demands, since the distinct fraction decays as 0.866^N. The largest shell we can diagonalise, $N = 16$, $w = 8$ ($|S_w| = 24\,310$, $\nu = 0.47$, 3278 ratios), goes further still: $\langle\tilde{r}\rangle = 0.347$, i.e. *below* Poisson. That is worth being careful about, because it rules out the two easy explanations rather than supporting them. It is not a superposition effect: superposing independent spectra approaches Poisson *from above* (2×Haar-O gives 0.418, 3× gives 0.396) and cannot go below. Nor is it unresolved near-degeneracy: among the distinct levels of every shell checked there is *not one* gap below a hundredth of the median gap (Table 5). The correct reading is §8.5: the level set is not a random point process of any kind, so it is not obliged to match Poisson either. Two controls at comparable dimension — the largest sector of `W54` (0.426, 0.433) and of `W57` (0.525, 0.527) at $N = 10, 12$, through the identical pipeline — confirm that the pipeline separates rules rather than saturating.

The one caveat worth stating: because the shells retain a large *unresolved* commutant even after reflection is taken out (the degeneracy fraction column), each measured $\langle\tilde{r}\rangle$ is a lower bound on the repulsion of a single irreducible block. That caveat cannot rescue chaos, though — (15) is an exact count, not a bound, and an exponentially degenerate spectrum is incompatible with RMT statistics whatever the blocks turn out to be. Identifying the commutant that (15) implies is the natural next step and is the sharpened form of open item 3.

8.5 One universal spectrum, and why no ensemble fits

The degeneracy law has a structural counterpart that explains the whole picture. For each N there is a single *universal* set Θ_N of 3^m (resp. $(3^m + (-1)^m)/2$) angles such that

- every wall shell's spectrum is a *subset* of Θ_N — verified to 10^{-9} for every shell of $N = 8, 10, 12$; and
- the *largest* shell, the one realising $D_{\max}$, realises *all* of Θ_N (100% coverage at $N = 8$, $w = 4$; $N = 10$, $w = 6$; $N = 12$, $w = 6$), while the small shells realise as little as 1.8%.

So the spectral problem of C150 is not "diagonalise 2^N states" but "find 3^m angles, then decide which shell picks which of them with what multiplicity". A level set generated by a closed-form alphabet of that kind is not a random point process, and this is the precise sense in which *no* standard ensemble applies: $\beta = 2$ (GUE, CUE) is excluded by time reversal; $\beta = 1$ (COE, Haar-O) is excluded by the exponential degeneracy and the closed-form count; and Poisson is excluded too, since a Poisson process does not put its points on a fixed 3^m-element alphabet, and at the largest accessible shell the observed $\langle\tilde{r}\rangle$ sits below the Poisson value with no near-degeneracies available to explain it away. C150 is spectrally *structured*, not random — which is exactly what one expects of an exactly solvable automaton, and exactly what the closed-form sector solution of §4 should have led us to expect inside the sectors too.

N	$\|\Theta_N\|$	all shells $\subseteq \Theta_N$	largest w	covers all	min gap/median
8	81	✓	4	✓	0.285
	coverage: w=2: 40.7%, w=4: 100.0%, w=6: 80.2%, w=8: 11.1%				
10	243	✓	6	✓	0.136
	coverage: w=2: 21.0%, w=4: 86.8%, w=6: 100.0%, w=8: 53.9%, w=10: 4.5%				
12	729	✓	6	✓	0.066
	coverage: w=2: 10.0%, w=4: 64.9%, w=6: 100.0%, w=8: 91.2%, w=10: 32.0%, w=12: 1.8%				

Table 5: One universal spectrum per N. Every shell's distinct spectrum is a subset of Θ_N (whose size is (15)), the largest shell covers all of it, and among the distinct levels of a shell there is no gap below a hundredth of the median gap — so the sub-Poisson $\langle\tilde{r}\rangle$ of the biggest shells is a property of the structured alphabet, not of unresolved near-degeneracies.

case	block	dim	deg. frac.	$\langle\tilde{r}\rangle$	n
reference ensembles, identical pipeline ($D = 600$, 12 samples each)					
Poisson	—	—	—	0.3844 ± 0.0165	—
3×Haar O	—	—	—	0.3957 ± 0.0211	—
2×Haar O	—	—	—	0.4179 ± 0.0118	—
COE ($\beta = 1$)	—	—	—	0.5300 ± 0.0204	—
Haar $O(D)$	—	—	—	0.5997 ± 0.0140	—
CUE ($\beta = 2$)	—	—	—	0.6058 ± 0.0189	—
C150 `obc0` wall shells					
$N = 9$, $w = 4$	P=+1	110	0.45	0.5054	28
$N = 9$, $w = 4$	P=-1	100	0.40	0.5104	28
$N = 10$, $w = 4$	all	330	0.36	0.5643	103
$N = 10$, $w = 6$	all	462	0.47	0.5797	119
$N = 11$, $w = 4$	P=+1	255	0.41	0.5879	73
$N = 11$, $w = 4$	P=-1	240	0.38	0.5093	73
$N = 11$, $w = 6$	P=+1	472	0.61	0.5970	89
$N = 11$, $w = 6$	P=-1	452	0.60	0.5496	89
$N = 12$, $w = 4$	all	715	0.34	0.5644	234
$N = 12$, $w = 6$	all	1 716	0.58	0.5393	362
$N = 12$, $w = 8$	all	1 287	0.48	0.5554	330
$N = 12$, $w = 10$	all	286	0.19	0.4845	114
$N = 13$, $w = 4$	P=+1	511	0.37	0.4204	159
$N = 13$, $w = 4$	P=-1	490	0.34	0.4354	159
$N = 13$, $w = 6$	P=+1	1 519	0.64	0.3956	271
$N = 13$, $w = 6$	P=-1	1 484	0.63	0.3965	271
$N = 14$, $w = 4$	all	1 365	0.31	0.4668	467
$N = 14$, $w = 6$	all	5 005	0.59	0.4482	1027
$N = 14$, $w = 10$	all	3 003	0.46	0.4588	803
$N = 14$, $w = 12$	all	455	0.17	0.4425	187
$N = 15$, $w = 4$	P=+1	924	0.33	0.4358	306
$N = 15$, $w = 4$	P=-1	896	0.31	0.4280	306
$N = 15$, $w = 8$	P=+1	6 470	0.75	0.3659	818
$N = 15$, $w = 8$	P=-1	6 400	0.74	0.3749	818
$N = 16$, $w = 4$	all	2 380	0.29	0.4834	846
$N = 16$, $w = 8$	all	24 310	0.73	0.3470	3278
$N = 16$, $w = 14$	all	680	0.15	0.5109	286
$N = 17$, $w = 4$	P=+1	1 548	0.30	0.4074	538
$N = 17$, $w = 4$	P=-1	1 512	0.29	0.3999	538
$N = 18$, $w = 16$	all	969	0.14	0.4005	415
control: other unitary rules, largest sector, small N					
`W54`, $N = 10$	—	1 023	0.03	0.4257	494
`W54`, $N = 12$	—	4 095	0.02	0.4326	2014
`W57`, $N = 10$	—	1 024	0.00	0.5245	508
`W57`, $N = 12$	—	4 096	0.00	0.5272	2044

Table 6: Spacing-ratio statistic. References are generated and pushed through the same pipeline (half circle, degeneracies collapsed). "deg. frac." is the fraction of the block's levels removed as exact degeneracies — a measure of how much commutant is still unresolved, and of how far $\langle\tilde{r}\rangle$ is biased low. $\nu = w/(N+1)$ is the wall filling; only $\nu \approx \frac{1}{2}$ is a many-body question.

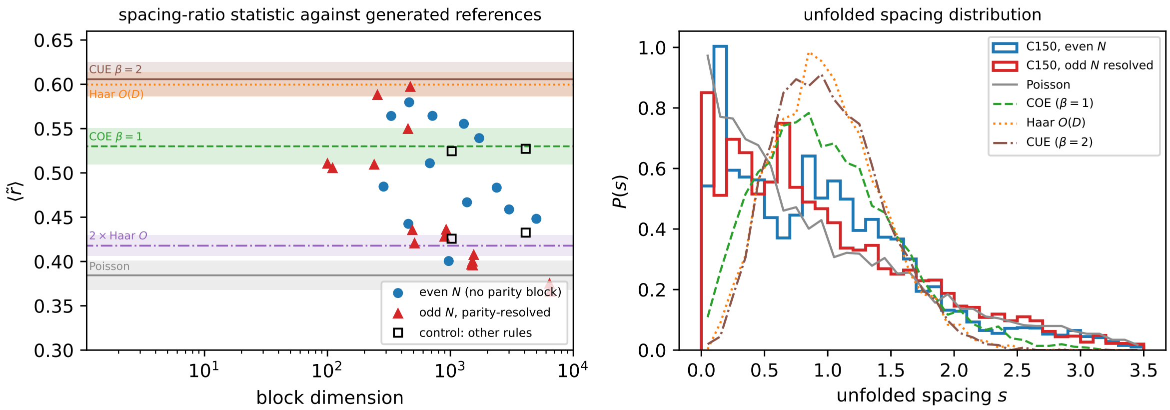

Figure 3: Left: $\langle \tilde{r} \rangle$ per symmetry block against block dimension, with the generated reference ensembles as bands ($\pm 1\sigma$ over the samples). Right: the unfolded spacing distribution, C150 pooled against the same references.

9 Cross-check against the `HSF` eigendata

Everything above was produced by one code base. The project also contains a pre-existing, independent Julia implementation (`HSF/`) with stored eigendata, which lets the central claims be tested against numbers this report did not produce.

The two codes build the same operator. `HSF` numbers the sixteen unitary rules $0 \ldots 15$ by the bits $(r_{11} r_{10} r_{01} r_{00})$ with $1 = H$, so `HSF` rule 6 is Wolfram 150. Reading `HSF/src/model.jl`: `binary_coefficients(6)` = `"0110"` gives $(1,1) \mapsto \mathbb{1}$, $(1,0) \mapsto H$, $(0,1) \mapsto H$, $(0,0) \mapsto \mathbb{1}$, i.e. our (`I`, `V`, `V`, `I`); the two boundary gates take the missing neighbour from the c_3/c_4 and c_2/c_4 branches, i.e. read it as 0, which is our `obc0`; and `U_full` = `U_even` · `U_odd` with `U_odd` collecting Julia sites $1, 3, 5, \ldots$ (zero-based $0, 2, 4, \ldots$, our *even* sublattice) applies the even sublattice first, as we do. Two conventions differ and neither can affect a spectrum: `HSF` puts qubit 1 in the most significant tensor factor where we put site 0 in the least significant bit (a bit reversal, i.e. a permutation similarity), and it stores $\theta = -\arg \lambda$ where we use $+\arg \lambda$. The second is moot because the spectrum is closed under conjugation — verified directly, the two multisets agree to 10^{-15}.

The stored files are flat arrays (`sector_id`, `theta`, `entropy`), one row per eigenstate of *every* Krylov sector, for $N = 8 \ldots 14$. Note that `HSF` finds its sectors from $|U_{yx}| > 10^{-10}$ on the assembled sparse matrix, so the agreement below also re-confirms the R1 finding that no one-cycle amplitude cancels.

Result: every claim survives. Table 7 reports the comparison. On all seven lengths, with no free parameters:

- the sector sizes equal $\binom{N+1}{w}$ for even w (§4);
- $\dim \mathrm{Fix}(U_w)$, read off as the number of stored $\theta = 0$ per sector, equals $\binom{\lceil N/2 \rceil}{w/2}$ (§6);
- the number of distinct quasienergies equals (15) *exactly* — $81, 121, 243, 365, 729, 1093, 2187$ for $N = 8 \ldots 14$. The last of these is the value (15) predicted out of sample, here reproduced by a different language, a different sparse assembly, a different basis ordering and a different sign convention;

- every sector's spectrum is a subset of the universal set Θ_N and the largest sector realises all of it (§8.5), with the smallest sectors realising as little as a few percent;
- the spacing-ratio statistic computed from *their* quasienergies with our pipeline reproduces our own per-shell values (e.g. $N = 10$: 0.5797; $N = 12$: 0.5393; $N = 13$: 0.3674) — and it must, since pooling all sectors gives the same distinct set as the largest sector alone.

N	states	sectors	#distinct	(15)	min cover	deg. states	max mult	$\langle\tilde{r}\rangle$	all checks
8	256	5	81	81	1.2%	0.43	6	0.4847	✓
9	512	6	121	121	0.8%	0.53	10	0.4000	✓
10	1 024	6	243	243	0.4%	0.56	10	0.5797	✓
11	2 048	7	365	365	0.3%	0.67	20	0.4335	✓
12	4 096	7	729	729	0.1%	0.66	20	0.5393	✓
13	8 192	8	1 093	1 093	0.1%	0.73	35	0.3674	✓
14	16 384	8	2 187	2 187	0.0%	0.74	35	0.4417	✓

Table 7: C150 against the independent `HSF` Julia eigendata, $N = 8 \dots 14$, gate H, `HSF` rule $6 =$ Wolfram 150. "deg. states" is the fraction of stored eigenstates lying in a *within-sector* degenerate eigenspace, and "max mult" the largest such multiplicity.

A caveat this raises for the stored entanglement entropies. The same degeneracy that makes C150 non-chaotic has a practical consequence for the `entropy` column. `HSF` diagonalises each sector block and records the half-chain entropy of each returned eigenvector; but on a degenerate eigenvalue the eigensolver returns an *arbitrary* orthonormal basis of the eigenspace, and the entanglement entropy is not constant on that basis. Rotating within a single eigenspace changes the stored entropies by up to 0.27 nats at $N = 8, 10$, while on nondegenerate eigenvalues the same test gives a spread of exactly 0 (`quantum.rule150_spectra.eigenspace_ee_spread`). This is not a defect of the `HSF` code, which does exactly what it says — it is a property of the spectrum, and for this rule it is the *typical* case rather than an edge case: the fraction of stored eigenstates sitting in a degenerate eigenspace rises from 0.43 at $N = 8$ to 0.74 at $N = 14$, with multiplicities up to 35 (Table 7). For rule 6 with the H gate, per-eigenstate entanglement entropies are therefore only well defined on the nondegenerate minority; a basis-independent replacement (for instance the eigenspace-averaged purity, or restricting the statistics to simple eigenvalues) would be needed before reading the $N = 13$ entropy histograms as physical. The mean stored entropy is in any case sub-thermal throughout, 0.50–0.61 of $\ln 2 \cdot \lfloor N/2 \rfloor$, which is the direction the rest of §8 would lead one to expect.

10 Area law versus volume law: what the sector decides

Figure 3a of `QCA_Circuits` starts C150 from three product states — a single excitation $|0\dots010\dots0\rangle$, the Néel state $|0101\dots\rangle$, and the pair state $|001100\dots\rangle$ — and finds that the same rule supports both area-law and volume-law growth. The natural objection is that the sectors are binomially large, $|S_w| = \binom{N+1}{w}$, so any state ought to have room to become volume-law entangled. That objection has a clean resolution, and it is a sector statement — but not one about $|S_w|$.

10.1 The three states are in three very different sectors

Translating each into wall language (§3):

state	configuration	w	$\lvert S_w\rvert = \binom{N+1}{w}$
single excitation	$0\dots010\dots0$	2	$\frac{1}{2}N(N{+}1)$, *polynomial*
Néel	$0101\dots$	N (N even)	$N+1$, *linear*
pair	$001100\dots$	$\approx N/2$	$\binom{N+1}{N/2}$, *exponential*

The premise fails because $\binom{N+1}{w}$ is exponentially large *only* near $w = (N+1)/2$; at both ends of the binomial row it is small. The Néel state is the striking case: it looks like the most excited product state in the chain, and in wall language it is a **one-hole** state — a wall on every bond but one. Walls are hard core (§3), so nothing can move except at that single hole: the Néel state is *jammed*, its sector has dimension $N+1$, and $S \le \ln(N+1)$ for all time. The single excitation is the mirror case, only two walls in an otherwise empty bond lattice — a few-body problem. Only the pair state sits at finite wall density, where the walls are mobile and the sector is exponentially large.

10.2 The exact kinematic ceiling

The sharp bound is not $\ln |S_w|$ but the bipartite Schmidt rank. Cut between sites $c-1$ and c. Because the bond map is a prefix XOR, x_A (sites $0..c{-}1$) determines the c bonds $b_0 \dots b_{c-1}$ and x_B determines the $N-c$ bonds $b_{c+1} \dots b_N$, with the cut bond b_c shared: $w = w_A + b_c + w_B$. Hence only those x_A occur whose internal wall count lies in an admissible window, and

$$\text{rank} \;\le\; \min\Bigg(|S_w|, \sum_{k=\max(0,\,w-1-(N-c))}^{\min(w,c)} \binom{c}{k}, \sum_{k=\max(0,\,w-1-c)}^{\min(w,\,N-c)} \binom{N-c}{k} \Bigg), \qquad S(t) \le \ln(\cdot) \;\; \forall t. \tag{17}$$

The *lower* limits are not decoration: the right side can absorb only so many walls, and dropping them over-counts badly at high filling — it is exactly what would make the Néel state look unconstrained. Equation (17) is not merely a bound but an identity for the rank ceiling: it equals the size of the compressed Schmidt matrix, checked against the actual dimensions for every w at $N = 6 \dots 14$ with no mismatch.

Evaluated on the three states (at $N = 26$, half-chain cut): the single excitation gets $\ln 92 = 4.52$, the Néel state $\ln 14 = 2.64$, and the pair state $\ln 8191 = 9.011$ — which is $\lfloor N/2 \rfloor \ln 2 = 9.011$ exactly, i.e. no kinematic constraint at all. The ceiling grows like $\ln N$ for the first two and like N for the third: **the sector alone already predicts area law, area law, volume law**, which is what Fig. 3a shows.

10.3 What the dynamics adds, and where the sector is not enough

Measured plateaus (time average over $t \in [100, 200]$) confirm and sharpen this:

N	10	14	18	22	26	30	34	38
single, S_{plateau}	1.00	1.11	1.17	1.19	1.23	1.23	1.24	1.24
Néel, S_{plateau}	0.53	0.57	0.59	0.59	0.58	0.59	0.58	0.58

Both are *constant* in N — strict area law, in fact flatter than their own $\ln N$ ceilings. The pair state instead grows, $1.29, 1.59, 2.12, 2.45$ at $N = 8, 10, 12, 14$, and since its ceiling *is* the volume-law value the growth is linear in N with a reduced coefficient.

That reduction is the dynamical part. Scanning all shells (Fig. 4, right) gives an entanglement *dome*: S_{plateau} peaks at half filling and falls to zero at both ends, while the *fraction* of the ceiling that is realised stays roughly flat, 0.21–0.52 across every w. So the shape of the dome is the ceiling's shape — the sector's shape — and the dynamics only supplies an $O(1)$ fill factor of about a half, which is the same sub-thermal fraction seen in the `HSF` entropies (§9) and is what one expects from the exponentially degenerate spectrum of §8: the state never explores its shell ergodically.

So: is the sector enough? For Fig. 3a, yes; in general, no.

- **For the growth type and its N-scaling: yes** — via (17), not via $|S_w|$. Knowing w fixes the ceiling exactly, and the ceiling already separates $\ln N$ from N.

- **For product initial states, the sector nearly fixes the value too.** Six random basis states of the $N = 14$, $w = 6$ shell give plateaus 2.26–2.39, a spread of about $\pm 3\%$; at $N = 12$, 1.84–2.13. Since Fig. 3a uses product states, its three curves are determined by three numbers w.

- **Over all states of a shell: no.** The stationary states of the same shell — $\dim \mathrm{Fix}(U_w) = \binom{\lceil N/2 \rceil}{w/2}$ of them, 35 at $N = 14$, $w = 6$ — have $S(t)$ *constant in time* to 10^{-15}, at a value *above* the product-state plateau (3.47–3.62 against 2.41). Inside one volume-law-capable sector there is therefore a 35-dimensional family with exactly zero entanglement growth, and no sector label distinguishes it. That is the same dark-state space as §6, seen dynamically.

The one-sentence version: *the sector fixes the ceiling and hence the law; the dynamics fixes the prefactor; and only non-product states inside a sector can violate the sector's expectation, by not moving at all.*

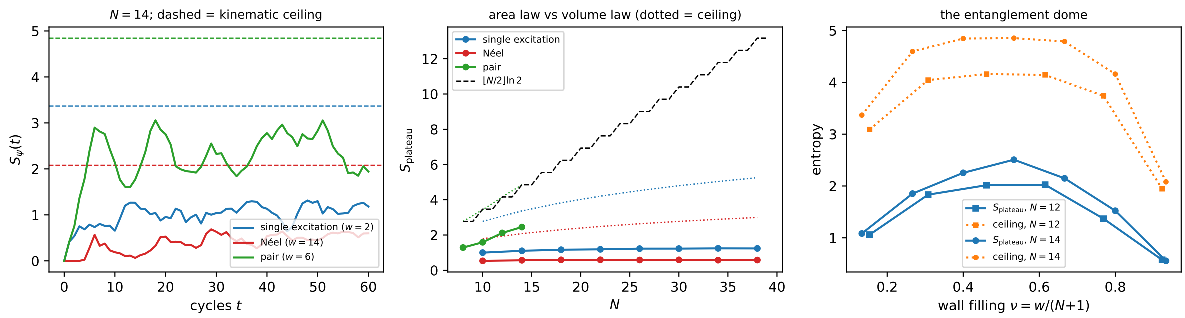

Figure 4: Left: $S_\psi(t)$ for the three Fig. 3a states at $N = 14$, with the kinematic ceilings (17) as dashed lines. Middle: the plateau against N — single excitation and Néel are flat (area law), the pair state grows (volume law); the dotted line is $\lfloor N/2 \rfloor \ln 2$. Right: the entanglement dome, S_{plateau} and its ceiling against wall filling $\nu = w/(N{+}1)$, with the realised fraction of the ceiling roughly flat.

11 The whole computational basis at once, and why no random state predicts it

§10 took three initial states. This section takes *all* of them: $N = 11$, `obc0`, the half-chain cut between sites 4 and 5, and every one of the $2^{11} = 2048$ computational basis states evolved to $t = 4000$. Each is a product state, so $S(0) = 0$ for all 2048; each lies in exactly one shell, of sizes $1, 66, 495, 924, 495, 66, 1$ (the even-weight entries of Pascal's row 12, §4). Since the dynamics is block diagonal, the trajectories of *all* $|S_w|$ basis states of a shell are the columns of U_w^t, so one matrix power per time step delivers the entire shell — the census costs 309 s, not 2048 separate runs.

11.1 There is no plateau: what "saturation" has to mean here

For a chaotic Floquet circuit one evolves to a short time, sees $S(t)$ flatten, and reads off the plateau. C150 does not do that, and §8 says why: with only $3^{\lceil N/2 \rceil}$ distinct eigenvalues, and every phase an integer combination $\theta = \sum_i \varepsilon_i \varphi_i$ of $m = \lceil N/2 \rceil = 6$ fundamentals, the orbit is quasi-periodic on a 6-torus. It never relaxes; it keeps wandering, with $\langle \sigma_t(S) \rangle \approx 0.35$ in the largest shell forever.

So "saturation value" can only mean the *infinite-time average*, and it must be measured with a window long enough for a 6-frequency quasi-periodic average to converge. It converges slowly: the cumulative mean of the $w = 6$ shell reads 1.740, 1.802, 1.785, 1.820, 1.819, 1.821 at horizons $t \le 100, 200, 400, 800, 1600, 4000$. A short $t \in [200, 400]$ window — the kind one would use for a chaotic circuit — returns 1.784, low by 2%. All numbers below use $t \in [200, 4000]$, where the eight non-overlapping block means scatter by only 0.013.

11.2 The census

w	$\lvert S_w \rvert$	$\overline{S}$	σ_w	min	max	ceiling	frac	Haar$_w$	deph.
0	1	0.0000	0.0000	0.000	0.000	0.000	0.000	0.000	—
2	66	0.9802	0.0368	0.852	1.023	2.773	0.354	1.488	1.454
4	495	1.6057	0.0768	1.237	1.718	3.434	0.468	2.716	2.640
6	924	1.8216	0.0930	1.276	1.968	3.466	0.526	3.138	3.054
8	495	1.6057	0.0768	1.237	1.718	3.434	0.468	2.717	—
10	66	0.9802	0.0368	0.852	1.023	2.773	0.354	1.485	—
12	1	0.0000	0.0000	0.000	0.000	0.000	0.000	0.000	—
all	2048	**1.6612**	0.2292	0.000	1.968	—	—	2.825	—

Reading it:

- **The global average is** $\overline{S} = 1.6612$ (median 1.6710, s.d. 0.2292, range $[0, 1.9681]$), against a volume law of $5 \ln 2 = 3.4657$. The basis average is 48% of the volume law.
- **Two shells are frozen.** $w = 0$ is $|0 \cdots 0\rangle$ and $w = 12$ is the Néel state $|10101010101\rangle$ — at odd N with `obc0` every one of the 12 bonds carries a wall, the hard-core walls have no hole to move into, and $S(t) \equiv 0$ exactly, forever. That is stronger than the area law of §10: at $N = 11$ the Néel state is not slow, it is *stationary*.
- **The partner shells are indistinguishable.** w and $12 - w$ agree to every digit shown — means, s.d.s, extrema. §8 found the partner shells spectrally identical at odd N (1.2×10^{-14}); the census shows the agreement extends to the entanglement across the cut. The unidentified intertwiner of open item 4 therefore commutes with the half-chain cut as well as with U, which is a sharp extra constraint on what it can be.
- **The realised fraction of the ceiling rises with the shell** (0.354, 0.468, 0.526), consistent with the dome of §10.

11.3 Inside one shell: two exact involutions, and an isolated spike

The $w = 6$ histogram in Fig. 5 is visibly *bimodal*: a broad bulk over $[1.75, 1.92]$ and a narrow, well-separated spike at $[1.950, 1.968]$, with an empty gap between. Both features have exact explanations, and neither is about the sector label.

The wall complement, and why every value is doubly degenerate. Let A be the even-site mask, $A = \sum_{i \text{ even}} 2^i$. In bond variables $x \mapsto x \oplus A$ complements *every* bond $b_j \mapsto 1 \oplus b_j$, so it maps the shell w to the shell $N + 1 - w$. It is an **exact symmetry of the saturation entropy**: over all 2048 states,

$$\left| \overline{S}(x) - \overline{S}(x \oplus A) \right| = 0.0 \times 10^0 \quad \text{(to machine precision).} \tag{18}$$

This is why the partner shells of the census table agree to every digit, and why in the self-partner shell $w = (N + 1)/2 = 6$ — where $x \oplus A$ acts *within* the shell — all 924 values come in 462 exactly degenerate pairs.

The point worth flagging: this is the very map of open item 4, the bond particle-hole map, which does *not* commute with U (checked $N = 3 \dots 9$). So it is not a symmetry of the dynamics, yet it is an exact symmetry of the time-averaged entanglement. Whatever the missing intertwiner is, it must reproduce $\oplus A$ at the level of the diagonal ensemble.

The spike: fixed points of $R = P \circ (\oplus A)$. Compose the wall complement with the spatial reflection P. The states fixed by R satisfy $Px = x \oplus A$, which in bond variables reads simply

$$b_{N-j} \;=\; 1 \oplus b_j \qquad \text{for every } j, \tag{19}$$

that is, *each mirror pair of bonds carries exactly one wall.* With $N + 1$ bonds in $(N + 1)/2$ mirror pairs, (19) forces $w = (N + 1)/2$ automatically — R can only have fixed points in the self-partner shell, and only when $(N + 1)/2$ is even, i.e. when $N \equiv 3 \pmod 4$. Counting on the two sublattices (the odd sites must form a palindrome, the even sites an anti-palindrome) gives

$$\#\{x : Px = x \oplus A\} = \begin{cases} 4^{(N+1)/4}, & N \equiv 3 \pmod 4, \\ 0, & \text{otherwise,} \end{cases} \tag{20}$$

verified for $N = 3, 7, 9, 11, 13, 15, 19, 23$ $(4, 16, 0, 64, 0, 256, 1024, 4096)$.

At $N = 11$ that is 64 states, and **they are exactly the spike**: the set of 64 largest $\overline{S}$ in the shell coincides with the fixed-point set of R, with

$$\overline{S} \in [1.9501, 1.9681] \text{ on the 64}, \qquad \overline{S} \in [1.2762, 1.9185] \text{ on the other 860},$$

a clean gap of 0.0316 — against a typical level spacing of 5×10^{-4} inside the spike. The mechanism is out-of-sample confirmed at $N = 7$, where (20) predicts 16 states in the $w = 4$ shell and they again top the shell with an empty gap (0.045); and at $N = 9$ and $N = 13$, where $(N + 1)/2$ is odd, the spike is predicted to be absent and is.

So the second peak is a *parity effect in* N, invisible in the sector label: it exists only for $N \equiv 3 \pmod 4$, only in the middle shell, and it consists of the wall configurations that are maximally balanced about the cut in the strong sense (19) — exactly one wall in each mirror pair of bonds, hence complementary wall content in the two halves.

And the low tail. The other tail, down to $\overline{S} = 1.2762$, is largely the 20 states that are genuinely reflection symmetric, $Px = x$. Those lie in a single reflection sector, so they populate only 71 of the shell's 183 distinct eigenphases — a quantum Krylov dimension 39% of the generic one — and they equilibrate to a visibly lower entropy (mean 1.674). Note $20 = \binom{6}{3} = \dim \mathrm{Fix}(U_w)$, which is suggestive but is a different subspace; the coincidence is not explained here.

11.4 How much does the sector fix?

Cluster the 2048 values by w and split the variance the usual one-way way, $\sigma^2_{\text{tot}} = \sigma^2_{\text{within}} + \sigma^2_{\text{between}}$:

$$\sigma^2_{\text{within}} = 0.0068, \qquad \sigma^2_{\text{between}} = 0.0457, \qquad \eta^2 = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{tot}}} = 0.870. \tag{21}$$

Knowing nothing but the wall number accounts for 87% of the variance of the saturation entropy across the whole Hilbert space. The residual 13% is real and not a measurement artifact: averaging each state over two *disjoint* halves of the window gives half-to-half correlation 0.9986 ($w = 4$) and 0.9988 ($w = 6$), with an implied noise level of 0.0024 and 0.0028 against spreads of 0.0768 and 0.0930. The intra-cluster spread is a genuine, reproducible property of each individual basis state, some $30\times$ above the noise floor.

This sharpens §10: the sector fixes the growth *type* exactly, the value to about $\pm 5\%$, and no more.

11.5 Could a random state have told us instead?

Three reference ensembles, in increasing order of how much they know. All are computed through the same entropy code as the census.

1. **Haar on the full 2^N space** — knows nothing. Real states give 3.2098 ± 0.0134; Page's formula for complex states gives $\sum_{k=d_B+1}^{d_A d_B} k^{-1} - (d_A - 1)/2d_B = 3.2160$ at $(d_A, d_B) = (32, 64)$. Measured: 1.6612. **Wrong by a factor 1.93.**

2. **Haar within the shell S_w**, weighted by shell size — knows the sector, and therefore knows everything §10 established. Gives 2.8248. Still **wrong by** 70%, and per shell it overshoots by 0.50, 1.11, 1.32 nats.

3. **Random phases on the exact spectrum** — the sharpest random-state model there is. Keep the exact initial basis state and the exact eigenspace decomposition, and replace only the coherent phases $\theta_\lambda t$ by independent uniform ones. This is what a diagonal-ensemble or "maximal dephasing" argument predicts, and it is exact whenever the eigenphases are rationally independent. It gives 1.454, 2.640, 3.055 for $w = 2, 4, 6$ against measured 0.980, 1.606, 1.822: **still wrong by** 50–70%**, and barely closer than plain Haar** (3.055 vs 3.138 at $w = 6$).

Item 3 is the informative failure, and it is a direct corollary of (15). The $w = 6$ shell carries 182 distinct positive eigenphases — exactly the $(3^6+1)/2 = 365$ distinct values of the full space, minus $\theta = 0$, halved, so the largest shell again realises the whole universal spectrum Θ_N of §8.5. But those 182 phases are built from only $m = 6$ independent fundamentals. The dephasing model lets 182 phases wander independently; the true dynamics moves them on a 6-dimensional torus. Filling a 182-torus instead of a 6-torus is precisely the overestimate.

The control confirms the frame is exact rather than the model being mis-implemented: feeding the *coherent* phases $\alpha_\theta = \theta t$ through the identical machinery reproduces the directly measured time average to four decimals (1.3901 against 1.3901, at $N = 9$, $w = 4$). Only the independence assumption is wrong.

11.6 Why the first-moment argument cannot help, ever

There is a reason to expect a random state to work, and it is worth stating because it is exactly right and exactly useless. The computational basis is an **exact 1-design**: $\mathbb{E}_x |x\rangle\langle x| = \mathbb{1}/2^N$. Unitary evolution preserves that, so at *every* time

$$\mathbb{E}_x\big[\rho_A(t)\big] = \mathrm{Tr}_B\, U^t \frac{\mathbb{1}}{2^N} U^{-t} = \frac{\mathbb{1}_A}{d_A}, \tag{22}$$

verified here to machine precision (0, 2.7×10^{-19}, 6.9×10^{-18}, 4.9×10^{-17} at $t = 0, 1, 7, 33$). Two consequences:

- **For any linear observable the question is trivial and the answer is static.** The basis average of $\langle O\rangle$ equals the Haar average equals $\mathrm{Tr}\, O/2^N$, exactly, and does not move under the dynamics at all. Had the quantity of interest been an observable, no simulation would have been needed.

- **For S the same input yields nothing.** S is concave, so (22) gives $\mathbb{E}_x[S] \le S(\mathbb{E}_x[\rho_A]) = \ln d_A = 3.4657$ — a bound that is true at every time, including $t = 0$ where all 2048 entropies are exactly 0. Meanwhile $\mathbb{E}_x[S]$ runs $0 \to 0.208 \to 1.424 \to 1.752$ over the same $t = 0, 1, 7, 33$. The first moment of ρ is constant while the answer moves across most of its range.

So the answer to "is S not an observable the obstruction?" is: that is the right diagnosis but not quite the right mechanism. Non-linearity is not fatal by itself — Page's formula computes $\mathbb{E}_{\text{Haar}}[S]$ exactly, nonlinear or not. What is fatal is that S is sensitive to *all* moments of the ensemble, and the computational basis matches Haar only at the first. It then takes two further pieces of physics, neither visible to any random-state model, to close the gap: the block structure ($3.21 \to 2.82$) and the non-ergodicity of a spectrum with 3^m distinct eigenvalues on 2^N states ($2.82 \to 1.66$).

And the spread is worse than the mean. Even granting a fitted offset, the random-state route cannot deliver the intra-cluster variance, which is the second half of the question. Concentration of measure makes Haar's own spread shrink with the shell dimension, and the measured spread does not:

w	$\lvert S_w\rvert$	measured σ_w	Haar σ_w
2	66	0.0368	0.0888
4	495	0.0768	0.0299
6	924	0.0930	0.0225

Haar is wrong in *both directions*: it over-predicts the spread by $2.4\times$ in the small shell and under-predicts it by $4.1\times$ in the large one. In the Haar model the intra-cluster variance is a finite-size artifact that vanishes as $D \to \infty$; here it grows with D, because the basis states of one shell are systematically inequivalent — they differ in how their wall configuration sits relative to the cut. No random-state ensemble, Haar or dephased, contains that information.

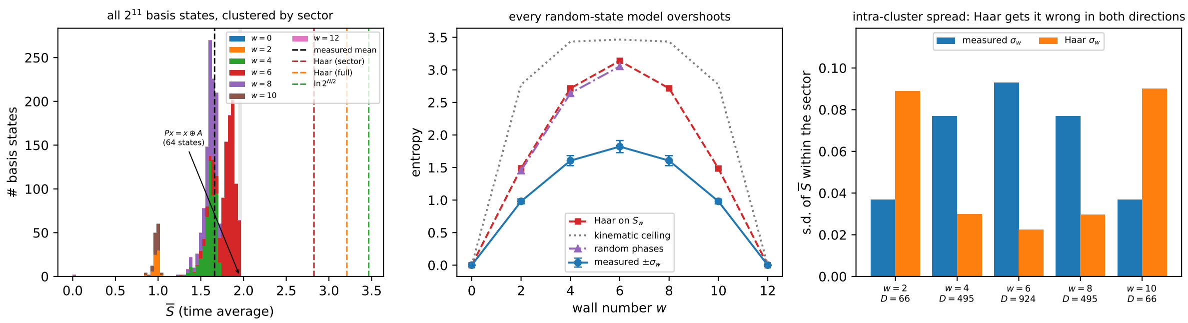

Figure 5: The $N = 11$ census. Left: the distribution of the 2048 saturation values, stacked by shell, with the measured mean and the three random-state predictions. The sector labels are visible as separated lobes — that separation is (21). The $w = 6$ shell is itself bimodal: the shaded band is the isolated spike of 64 states fixed by $R = P \circ (\oplus A)$ (§11.3). Middle: per-shell mean $\pm\sigma_w$ against the kinematic ceiling and the two random-state references; every random-state model lies above the data at every w. Right: the intra-cluster spread against Haar's own, showing that Haar errs in opposite directions for small and large shells.

12 Open items

1. **A w-resolved version of (15), and a proof.** The full-space count is exact and now out-of-sample tested, but it is not derived, and the per-shell count is not yet in closed form. The base 3 per even site points at the dihedral structure (13) as the route.

2. **A derivation of (8).** The identity $\text{Fix}(U) = \text{Fix}(L_A) \cap \text{Fix}(L_B)$ is now *explained* by the two-involution structure (13) together with the numerical emptiness of the $(-1, -1)$ dihedral sector, which is the one remaining gap. The w-resolved law $\dim \text{Fix}(U_w) = \binom{\lceil N/2 \rceil}{w/2}$

is still only observed; the binomial suggests the dark states are labelled by choosing $w/2$ of the $\lceil N/2 \rceil$ even sites.

3. **The commutant.** Beyond the $+1$ eigenvalue the spectrum carries exact multiplicities $2, 3, 4, 6, 10, 15$ (e.g. $N = 8$, $w = 4$: $\{1{:}48,\ 2{:}24,\ 3{:}8,\ 6{:}1\}$), not explained by reflection, which is abelian. Equation (15) now says how much commutant there must be: with only $3^{\lceil N/2 \rceil}$ distinct eigenvalues on 2^N states, the commutant dimension $\sum_\lambda m_\lambda^2$ is exponentially large. Identifying its generators would prove (15) and turn the block-resolved statistics of §8 from a lower bound into an exact result.

4. **Level statistics for the partner shells.** For odd N the shells w and $N+1-w$ have identical spectra to 1.2×10^{-14}, but the natural candidate intertwiner — the bond particle-hole map, which in spin variables is $x \mapsto x \oplus A$ with A the alternating mask — does *not* commute with U (checked for $N = 3 \ldots 9$). The intertwiner that does is unidentified. §11 adds a constraint that should narrow the search considerably: the partner shells agree not only spectrally but in their entire half-chain entanglement census — mean, spread and extrema, to every digit — so whatever the intertwiner is, it commutes with the bipartition as well as with U.

5. **The ring.** §4 solves `pbc` kinematically, but the quantum layer of §6–§7 is done for `obc0` only. On the ring the wall map is 2-to-1, so the bond representation acquires a $\mathbb{Z}_2$ gauge redundancy and (12) needs a flux sector; that is the next natural step.

6. **Higher N at the assumption-free frontier.** The streamed engine costs a factor ≈ 2.8 per site ($N = 20$: 1.1 h; $N = 21$: 3.6 h; $N = 22$: 10.2 h), so $N = 23$ is a $\approx$ 30-hour unit — launched, but outside the one-day budget of this report, and not needed for the closed form. The reduction frontier is instead bounded by the label array: $N = 33$ needs `uint64` and 68.7 GB, so the next step there is a machine with more memory, or a compact relabelling (the components are few, so a two-pass scheme with a byte-wide label after the first sweep would halve the requirement).

13 Reproduce

```
# closed forms, wall bijection, charge conservation, engine cross-check
python -m qca_fragmentation.c150 --verify

# assumption-free frontier (one unit; N=22 takes ~10 h)
python -m qca_fragmentation.run_rule --rule 150 --N 21-22 --bc obc0

# reduction frontier (checkpointed to analytics/c150_frontier.jsonl)
python -m qca_fragmentation.c150 --frontier 21-31 --bc obc0

# sector spectra, Krylov dimensions, Fix(U), freeness test
python -m qca_fragmentation.scaling.c150_report --quantum

# level statistics: references, shells, controls, distinct-eigenvalue law
python -m qca_fragmentation.quantum.rule150_levels --run

# full computational-basis entanglement census + random-state references
# (N=11: all 2048 states to t=4000; ~25 min.  Cap the BLAS threads --
#  the default oversubscribes and costs a factor 2.)
OMP_NUM_THREADS=4 OPENBLAS_NUM_THREADS=4 \
```

```
  python -m qca_fragmentation.quantum.rule150_basis_census --run

python -m qca_fragmentation.scaling.c150_report --tables --figures

# tests (126 of them)
cd code/qca_fragmentation/tests
python -m pytest test_flip_graph.py test_c150.py -q
```

Data: `results/150_obc0.jsonl` and `results/150_pbc.jsonl` (Tier 1a), `analytics/c150_frontier.jsonl` (reduction frontier, one JSON line per $(N, \mathtt{bc})$ with the full size multiset and the closed-form check), `analytics/c150_quantum.json` (spectral portraits, $\mathrm{Fix}(U)$, freeness), `analytics/c150_levels.json` (reference ensembles with their pooled spacing samples, per-block $\langle\tilde{r}\rangle$ and KS distances, controls, and the distinct-eigenvalue table), `analytics/c150_basis_census.json` (the per-state saturation values for all 2^{11} basis states with their shell labels, the per-shell and global statistics, the three random-state references, and the convergence, split-half and 1-design diagnostics).